

```
#include <iostream>
```

```
#include <fstream>
```

```
#include <iomanip>
```

```
#include <cstring>
```

```
Using namespace std;
```

```
Class Bank
```

```
{
```

```
Private:
```

```
    Int accNo;
```

```
    Char name[50];
```

```
    Char type[20];
```

```
    Float balance;
```

```
Public:
```

```
Void createAccount()
```

```
{
```

```
    Cout << "\nEnter Account Number : ";
```

```
    Cin >> accNo;
```

```
    Cin.ignore();
```

```
    Cout << "Enter Customer Name : ";
```

```
    Cin.getline(name, 50);
```

```

Cout << "Enter Account Type (Saving/Current) : ";
Cin.getline(type, 20);

Cout << "Enter Initial Deposit : ";
Cin >> balance;
}

```

```

Void showAccount()
{
    Cout << "\n-----";
    Cout << "\nAccount Number : " << accNo;
    Cout << "\nCustomer Name : " << name;
    Cout << "\nAccount Type : " << type;
    Cout << "\nBalance : " << balance;
    Cout << "\n-----\n";
}

```

```

Int getAccountNo()
{
    Return accNo;
}

```

```

Void deposit(float amount)
{
    Balance += amount;
}

```

```
}
```

```
Bool withdraw(float amount)
```

```
{
```

```
    If(amount > balance)
```

```
    {
```

```
        Return false;
```

```
    }
```

```
    Balance -= amount;
```

```
    Return true;
```

```
}
```

```
Float getBalance()
```

```
{
```

```
    Return balance;
```

```
}
```

```
Void updateDetails()
```

```
{
```

```
    Cin.ignore();
```

```
    Cout << "\nEnter New Customer Name : ";
```

```
    Cin.getline(name, 50);
```

```
    Cout << "Enter New Account Type : ";
```

```
        Cin.getline(type, 20);  
    }  
};
```

```
Void createNewAccount()
```

```
{
```

```
    Bank b;
```

```
    Ofstream file("bank.dat", ios::binary | ios::app);
```

```
    b.createAccount();
```

```
    file.write((char*)&b, sizeof(b));
```

```
    file.close();
```

```
    cout << "\nAccount Created Successfully.\n";
```

```
}
```

```
Void displayAllAccounts()
```

```
{
```

```
    Bank b;
```

```
    Ifstream file("bank.dat", ios::binary);
```

```
    If(!file)
```

```
{  
    Cout << "\nNo Records Found.\n";  
    Return;  
}
```

```
While(file.read((char*)&b, sizeof(b)))  
{  
    b.showAccount();  
}
```

```
File.close();  
}
```

```
Void searchAccount()
```

```
{  
    Int acc;  
    Bool found = false;  
    Bank b;
```

```
Cout << "\nEnter Account Number: ";  
Cin >> acc;
```

```
Ifstream file("bank.dat", ios::binary);
```

```
While(file.read((char*)&b, sizeof(b)))  
{  
    If(b.getAccountNo() == acc)
```

```
{  
    b.showAccount();  
    found = true;  
    break;  
}  
}
```

```
File.close();
```

```
If(!found)  
    Cout << "\nAccount Not Found.\n";  
}
```

```
Void depositMoney()
```

```
{  
    Int acc;  
    Float amount;  
    Bool found = false;  
    Bank b;  
  
    Cout << "\nEnter Account Number: ";  
    Cin >> acc;  
  
    Cout << "Enter Amount to Deposit: ";  
    Cin >> amount;
```

```
Fstream file("bank.dat", ios::binary | ios::in | ios::out);
```

```
While(file.read((char*)&b, sizeof(b)))
```

```
{
```

```
    If(b.getAccountNo() == acc)
```

```
    {
```

```
        b.deposit(amount);
```

```
        file.seekp(-(long)sizeof(b), ios::cur);
```

```
        file.write((char*)&b, sizeof(b));
```

```
        cout << "\nAmount Deposited Successfully.\n";
```

```
        cout << "Current Balance: " << b.getBalance() << endl;
```

```
        found = true;
```

```
        break;
```

```
    }
```

```
}
```

```
File.close();
```

```
If(!found)
```

```
    Cout << "\nAccount Not Found.\n";
```

```
}
```

```
Void withdrawMoney()
```

```

{
    Int acc;

    Float amount;

    Bool found = false;

    Bank b;


    Cout << "\nEnter Account Number: ";

    Cin >> acc;


    Cout << "Enter Amount to Withdraw: ";

    Cin >> amount;


    Fstream file("bank.dat", ios::binary | ios::in | ios::out);

    While(file.read((char*)&b, sizeof(b)))
    {
        If(b.getAccountNo() == acc)
        {
            If(b.withdraw(amount))
            {
                File.seekp(-(long)sizeof(b), ios::cur);

                File.write((char*)&b, sizeof(b));


                Cout << "\nWithdrawal Successful.\n";

                Cout << "Current Balance: " << b.getBalance() << endl;
            }
        }
    }
}

```



```
    Else
    {
        Cout << "\nInsufficient Balance.\n";
    }

    Found = true;
    Break;
}

File.close();

If(!found)
    Cout << "\nAccount Not Found.\n";
}

Void balanceEnquiry()
{
    Int acc;
    Bool found = false;
    Bank b;

    Cout << "\nEnter Account Number: ";
    Cin >> acc;

    Ifstream file("bank.dat", ios::binary);
```

```
While(file.read((char*)&b, sizeof(b)))
{
    If(b.getAccountNo() == acc)
    {
        Cout << "\nCurrent Balance: " << b.getBalance() << endl;
        Found = true;
        Break;
    }
}
```

```
File.close();
```

```
If(!found)
    Cout << "\nAccount Not Found.\n";
}
```

```
Void updateAccount()
```

```
{
    Int acc;
    Bool found = false;
    Bank b;
```

```
Cout << "\nEnter Account Number: ";
Cin >> acc;
```

```
Fstream file("bank.dat", ios::binary | ios::in | ios::out);
```

```
While(file.read((char*)&b, sizeof(b)))
```

```
{
```

```
    If(b.getAccountNo() == acc)
```

```
    {
```

```
        b.updateDetails();
```

```
        file.seekp(-(long)sizeof(b), ios::cur);
```

```
        file.write((char*)&b, sizeof(b));
```

```
        cout << "\nAccount Updated Successfully.\n";
```

```
        found = true;
```

```
        break;
```

```
    }
```

```
}
```

```
File.close();
```

```
If(!found)
```

```
    Cout << "\nAccount Not Found.\n";
```

```
}
```

```
Void deleteAccount()
```

```
{
```

```
Int acc;

Bool found = false;

Bank b;

Cout << "\nEnter Account Number: ";

Cin >> acc;

Ifstream file("bank.dat", ios::binary);
Ofstream temp("temp.dat", ios::binary);

While(file.read((char*)&b, sizeof(b)))
{
    If(b.getAccountNo() == acc)
    {
        Found = true;
    }
    Else
    {
        Temp.write((char*)&b, sizeof(b));
    }
}

File.close();

Temp.close();

Remove("bank.dat");
```

```
Rename("temp.dat", "bank.dat");
```

```
If(found)
```

```
    Cout << "\nAccount Deleted Successfully.\n";
```

```
Else
```

```
    Cout << "\nAccount Not Found.\n";
```

```
}
```

```
Int main()
```

```
{
```

```
    Int choice;
```

```
    Do
```

```
    {
```

```
        Cout << "\n===== BANK MANAGEMENT SYSTEM =====\n";
```

```
        Cout << "1. Create New Account\n";
```

```
        Cout << "2. Display All Accounts\n";
```

```
        Cout << "3. Search Account\n";
```

```
        Cout << "4. Deposit Money\n";
```

```
        Cout << "5. Withdraw Money\n";
```

```
        Cout << "6. Balance Enquiry\n";
```

```
        Cout << "7. Update Account\n";
```

```
        Cout << "8. Delete Account\n";
```

```
        Cout << "9. Exit\n";
```

```
        Cout << "\nEnter Your Choice: ";
```

```
Cin >> choice;
```

```
Switch(choice)
```

```
{
```

```
    Case 1: createNewAccount(); break;
```

```
    Case 2: displayAllAccounts(); break;
```

```
    Case 3: searchAccount(); break;
```

```
    Case 4: depositMoney(); break;
```

```
    Case 5: withdrawMoney(); break;
```

```
    Case 6: balanceEnquiry(); break;
```

```
    Case 7: updateAccount(); break;
```

```
    Case 8: deleteAccount(); break;
```

```
    Case 9: cout << "\nThank You!\n"; break;
```

```
    Default: cout << "\nInvalid Choice!\n";
```

```
}
```

```
} while(choice != 9);
```

```
Return 0;
```

```
}
```